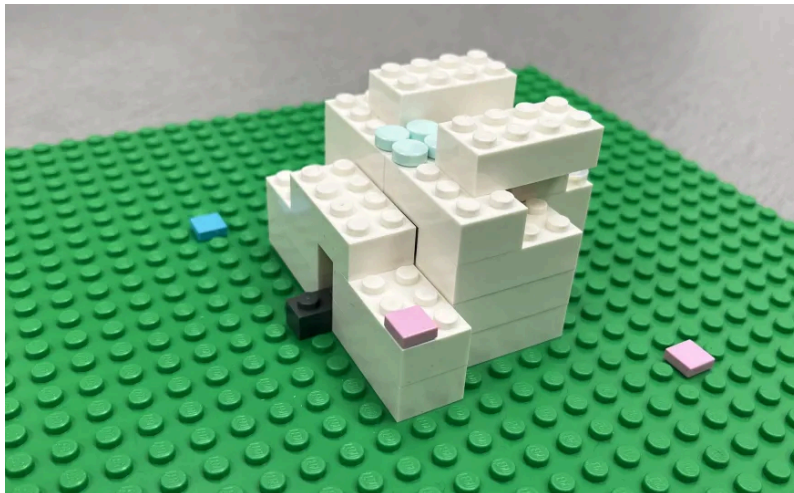


Further instructions:

To set up the environment for the project, we need to build a Lego structure in which the tumor is centrally located (blue dots in the image below) and is surrounded by big Lego blocks to represent the bone structure of the skull. We also place 1 blue square to represent the origin and 3 pink squares to represent the registration points. These points can be placed at any level and location. This Lego structure can be built in any way that the user wants but we need to make sure that we have the origin and 3 registration points minimum for the registration algorithm to work. Ensure to have the Lego board (image space) parallel to the robot's xy-plane for the current implementation of the project.



Descriptions of the tests that have been performed:

The first test we did was to make sure that point registration works correctly in C++. To verify this, we took the location of the origin point and 3 further registration points. We used the algorithm to convert those points from voxel to robot space. The next step was to provide user input for a location where we wanted to move the end-effector in voxel space. Once that has been done, we verified this test by checking if the robot's end-effector has moved to the correct location. We can confirm that this test works correctly as the robot moved to the correct location.

The next set of tests that will be discussed is whether a no-go zone blocks the needle trajectory or if the needle insertion works by going toward the tumor. These are the 2 main categories that we are testing out for. We conducted 4 tests each for the 2 categories in different positions based on the robot base frame and orientation.

The first position tested was towards the left-hand side of the robot's base and laying down the Lego brick structure flat on the table. In this position, we tested the 2 main categories that we need. In this position, the test for detecting if the no-go zone blocks the needle trajectory works as placing the needle in a path where a no-go zone produces the output "No Go" on the terminal. The next test in this position was to see if the needle moves towards the tumor based

on a correct trajectory. The main goal of this test is to see if the robot can move the needle into a correct orientation such that it finds the correct trajectory to move toward the tumor. This test works as we place the needle a few centimeters away from the tumor and at the end of the test, the needle is in contact with the tumor.

The second position tested is exactly the same as the first position except it was tested on the right-hand side of the robot's base. This position was done to make sure that the 2 main categories that we are testing for work in a different location from where all the trial and error were made by fixing the code for the first position. For this position, the 2 main categories that we are testing for work in the same way as described for the first position.

The third position that was tested is exactly the same as the first position except it was tested on a higher elevation. The Lego brick structure was not flat on the table but was a few centimeters higher than that. The reason for this test is to see if the no-go zones still work based on a different elevation because the no-go zones are based on a level-by-level basis in the voxel space (z-axis). Furthermore, this test will ensure that the no-go zones do not start from the same level as the robot's base frame for the z-axis once the voxel coordinates are converted to the robot's coordinates. For this position, the 2 main categories that we are testing for work in the same way as described for the first position.

The final position that we tested for is in which the Lego brick structure is not parallel to the table but is at an angle and is in the fake skull. The reason for this test is that we wanted to implement the project in a more realistic scenario. Thereby, having the Lego brick structure in a skull-like environment. The reason for the angle of the brick structure is that we wanted to place it in the center of the skull so that the structure does not move as it is no longer connected to the bigger Lego board. The testing works for one of the categories which is where the needle makes contact with the tumor for the correct orientation. The testing for the no-go zones in this position did not go as planned. The reason for this is that the no-go zones are on a level-by-level flat basis from the table.